

1. Why Testing & STLC

Introduction

Software Testing is the process of verifying and validating a software application to ensure it meets business requirements and works as expected.

Testing helps identify:

- defects
- bugs
- missing requirements
- performance issues

The main goal is to deliver a high-quality product to users.

Why Testing is Needed

Without testing:

- software may crash
- incorrect results may occur
- security issues may arise
- customer dissatisfaction increases

With testing:

- quality improves
 - defects are reduced
 - reliability increases
 - customer confidence improves
-

STLC (Software Testing Life Cycle)

STLC is a systematic process followed by testers to perform testing activities.

Phases of STLC

None

Requirement Analysis



Test Planning



Test Case Design



Test Environment Setup



Test Execution



Defect Reporting



Test Closure

POC Example

Use Case

Login System

Requirement:

- User should login using valid username and password.

Testing Activities:

- Analyze requirement
 - Create test cases
 - Execute test cases
 - Report bugs
 - Verify fixes
-

2. RTM (Requirement Traceability Matrix)

Introduction

RTM is a document that maps requirements with test cases.

It ensures:

- every requirement is tested
- no requirement is missed
- complete test coverage

Purpose

RTM helps:

- track requirements
- verify test coverage
- identify missing test cases

RTM Structure

Requirement ID	Requirement	Test Case ID	Status
R1	Login Functionality	TC01	Pass
R2	Forgot Password	TC02	Pass
R3	Registration	TC03	Fail

POC Example

Requirement

None
User should login successfully

Test Case

None
Enter valid username and password

RTM Entry

Requirement	Test Case
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3. Types of Testing

Introduction

Types of Testing are different methods used to validate software functionality and quality.

Functional Testing

Tests application functionality against requirements.

Example

Login Feature

Input:

None
Username = admin
Password = admin123

Expected:

None
Login Success

Unit Testing

Tests individual components.

Example

Testing a calculator addition method.

Integration Testing

Tests interaction between modules.

Example

Login Module + Database Connection

System Testing

Tests complete application.

Example

Testing complete food delivery application.

User Acceptance Testing (UAT)

Performed by end users.

Example

Client verifies all project requirements.

Regression Testing

Checks whether existing functionality is affected after changes.

Example

After adding payment module:

- login should still work
-

Smoke Testing

Checks basic application stability.

Example

- Login page opens
 - Dashboard loads
-

Sanity Testing

Checks specific functionality after bug fix.

Example

Verify password reset functionality after fix.

POC Example

Food Delivery Application

Testing Types:

None
Unit Testing
↓
Integration Testing
↓
System Testing
↓
UAT

4. Test Case Preparation and Template

Introduction

A Test Case is a document containing:

- test steps

- input data
- expected result
- actual result

Used to verify functionality.

Test Case Template

Test Case ID	TC_LOGIN_001
Module	Login
Test Scenario	Valid Login
Preconditions	User Registered
Test Steps	Enter credentials
Expected Result	Login Success
Actual Result	Login Success
Status	Pass

Test Case Design Techniques

Equivalence Partitioning

Divide inputs into valid and invalid classes.

Example:

Age Field:

None
18-60

Valid:

None
25

Invalid:

None
10
70

Boundary Value Analysis

Test boundary values.

Example:

Age Range:

None
18-60

Test:

None
17
18
19
59
60
61

POC Example

Password Validation

Requirement:

None

Password length 8-14 characters

Test Cases:

Input	Expected
7 chars	Invalid
8 chars	Valid
14 chars	Valid
15 chars	Invalid

5. Test Plan

Introduction

A Test Plan is a formal document describing:

- testing strategy
- scope
- resources
- schedule
- testing objectives

It acts as a roadmap for testing activities.

Purpose

Test Plan helps:

- organize testing
- define responsibilities
- estimate effort
- track progress

Components of Test Plan

Objective

What needs to be tested.

Example:

None

Validate Login Module

Scope

Features included in testing.

Example:

None

Login

Registration

Forgot Password

Testing Types

None

Functional Testing

Regression Testing

System Testing

Resources

None

1 Tester

1 Test Lead

Schedule

None

Test Case Design - 2 Days

Execution - 3 Days

Bug Verification - 1 Day

Test Plan Flow

None

Requirements



Test Plan



Test Cases



Execution



Bug Reporting



Closure Report

Real-Time POC

Project: Online Book Store

Requirement

Users should:

- Register
 - Login
 - Search Books
 - Add to Cart
 - Purchase Books
-

Testing Performed

STLC

Requirement Analysis → Planning → Test Cases → Execution

RTM

Mapped all requirements with test cases.

Types of Testing

- Functional Testing
- Integration Testing
- System Testing
- UAT

Test Cases

Created login, registration, cart, and payment test cases.

Test Plan

Defined:

- scope
- schedule
- resources
- testing strategy